

LLD-FM-001 FLEET MANAGEMENT PLATFORM

Low Level Design — Fleet Management Platform

LLD	FM-001	v1.0	EKS	CONFIDENTIAL
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EKS Microservices | Step Functions | IoT TwinMaker | Managed Grafana
SpaceNet-IT | HLD Ref: SPACENET-HLD-001 | June 2026

1. Scope & Purpose

This LLD defines the detailed design of the Fleet Management Platform. It covers the EKS microservice architecture, API design, satellite state machine, contact scheduling, telecommand uplink pipeline, digital twin integration, and NOC visualisation platform.

2. EKS Cluster Architecture

2.1 Cluster Configuration

EKS version	1.30 (managed upgrade)
Cluster endpoint	Private only — no public API server endpoint
CNI	AWS VPC CNI — native VPC IP addressing
Service mesh	AWS App Mesh (mTLS between microservices)
Ingress	AWS Load Balancer Controller — NLB per service
DNS	CoreDNS — cluster-internal service discovery
Storage	EBS CSI driver (gp3) — persistent volumes
Secrets	External Secrets Operator → AWS Secrets Manager
GitOps	Flux CD — reconciles from ECR + CodeCommit
Observability	AWS Distro for OpenTelemetry (ADOT) — traces + metrics
Node group	On-demand m6i.xlarge — system pool (min 3, max 6 per AZ)
Workload nodes	On-demand c6i.2xlarge — fleet services (min 3, max 30 HPA)

2.2 Namespace Layout

fleet-core	contact-scheduler, orbit-propagator, fleet-state-manager	Ingress from noc-ops only
telemetry	telemetry-ingestor, anomaly-detector	Ingress from Kinesis consumer + fleet-core
command	command-dispatcher	Ingress from noc-ops + Step Functions only
noc-ops	grafana, twinmaker-bridge, alerting	Ingress from ALB (operator IPs)
platform	External Secrets, ADOT collector, Flux CD	Ingress from cluster-admin only

3. Microservice Detailed Design

contact-scheduler	Python 3.12	3 (min)	500m / 1Gi	Schedule contacts, conflict resolution	Mission Planner	DynamoDB	EventBridge	API
telemetry-ingestor	Go 1.22	5 (min)	1000m / 2Gi	Consume Kinesis, validate packets	Kinesis Data Firehose	DynamoDB	API	SNS
command-dispatcher	Python 3.12	3 (min)	500m / 512Mi	Validate TC, encode CFDP, queue	API Gateway	SageMaker	API	DynamoDB
orbit-propagator	Python 3.12	3 (min)	500m / 1Gi	SGP4/SDP4 propagation, TLE fetch	Spacecrafting	API	API	API
anomaly-detector	Python 3.12	3 (min)	1000m / 2Gi	Threshold checks, ML anomaly	SageMaker	EventBridge	API	API
fleet-state-manager	Go 1.22	3 (min)	500m / 512Mi	State machine NOMINAL/SAFE/EMERGENCY	EventBridge	DynamoDB	API	SNS
contact-reporter	Python 3.12	2 (min)	250m / 512Mi	Post-contact summary, link budget	EventBridge	S3	API	SNS

4. Contact Scheduling Pipeline

4.1 Contact Request Flow

1	Mission Planner	POST /v1/contacts/request	satellite_id, TLE, priority, time_window
2	contact-scheduler	Fetch available Ground Station windows	AWS Ground Station ListAvailableContacts API
3	contact-scheduler	Run conflict resolution algorithm	Priority queue, existing schedule, elevation mask
4	contact-scheduler	Reserve contact	AWS Ground Station ReserveContact API
5	contact-scheduler	Write to DynamoDB	contacts table — status: SCHEDULED
6	orbit-propagator	Calculate AOS/LOS times	SGP4 propagation + site coordinates
7	EventBridge	Schedule pre-contact preparation	T-5 min rule → configure dataflow endpoint

4.2 Contact Execution State Machine (Step Functions)

PREPARING	T-5 min EventBridge	Start ECS dataflow endpoint, configure Ground Station	STARTING_AOS
WAITING_AOS	Contact start time	Monitor UDP packets on port 55888	IN_CONTACT (first packet) or MISSED
IN_CONTACT	UDP packets received	Stream VITA49 → CCSDS → Kinesis. Monitor post-pass	POST_PASS
POST_PASS	AOS/LOS end time	Flush frame queue, stop dataflow endpoint	REPORTING
REPORTING	Queue empty	Generate contact report, update DynamoDB, notify	COMPLETE
MISSED	No UDP after AOS+2min	Alert NOC, try secondary region antenna	COMPLETE

5. Telecommand Pipeline

1 — Submit	API Gateway + Lambda	Operator submits TC via authenticated REST API. JWT validated, APID checked against
2 — Validate	command-dispatcher EKS	Validate TC syntax, check satellite mode (no TC in SAFE_MODE without override), assign
3 — Dual Approval	Step Functions human task	Two NOC operators must approve critical commands (APID > 0x180). 15-minute timeout
4 — Encode	command-dispatcher	Encode as CCSDS TC Transfer Frame + CFDP PDU. Sign with KMS CMK
5 — Schedule	contact-scheduler	Queue for uplink at next available contact window. Return estimated uplink time to operator
6 — Uplink	AWS Ground Station	Transmit CLTU (Command Link Transmission Unit) during contact uplink window
7 — Acknowledge	telemetry-ingestor	Monitor VC0 HK stream for TC ACK/NAK packet (APID 0x103). Update DynamoDB comr

6. Satellite State Machine

NOMINAL	Normal operations	Boot or recovery from SAFE	Anomaly detection	All
DEGRADED	Reduced capability	Single subsystem fault	Fault cleared or escalation	Non-critical only
SAFE_MODE	Survival mode	Critical anomaly or cmd	Operator command + approval	Safe mode exit only
EMERGENCY	Critical — immediate action	Multiple failures	Recovery command	Emergency only
ECLIPSE	No solar power	Orbit calculation	Orbit calculation	All (battery check)
CONTACT	Ground link active	AOS event	LOS event	All

7. Digital Twin — IoT TwinMaker

Workspace	spacenet-constellation — one workspace per operator account
Entity model	Satellite entity → subsystem components (Power, Thermal, ADCS, Comms, OBC)
Ground station entity	AWS Ground Station site → contact history, link quality
Orbital component	orbit-propagator pushes lat/lon/alt/velocity every 10 seconds
Telemetry binding	Timestream datasource — live HK parameters bound to entity properties
Alert overlay	EventBridge rule → Lambda → TwinMaker update (anomaly flag on entity)
Scene	3D scene with satellite orbital tracks and ground station beam footprints
Grafana widget	TwinMaker Grafana plugin embedded in NOC dashboard

8. NOC Platform — Managed Grafana

Fleet Overview	Constellation map, state summary, active contacts, time to next contact	Time to next contact, time to next contact	5s
Satellite Detail	All HK parameters, power curve, thermal state, A/DCS attitude	Time to next contact, time to next contact	5s
Contact Schedule	Next 24h contacts, pass geometry, link budget	Link budget DB (via Lambda)	30s
Anomaly History	Alert timeline, parameter trends, threshold violations	Open violations	10s
Command Log	TC history, uplink queue, ACK/NAK status	DynamoDB	10s
System Health	EKS pods, Kinesis lag, Lambda errors, CloudWatch	CloudWatch Station API	30s

9. API Design

/v1/contacts	GET	JWT	List scheduled contacts — filter by satellite, time, status
/v1/contacts/request	POST	JWT	Request a new contact window
/v1/contacts/{id}/cancel	DELETE	JWT + MFA	Cancel a scheduled contact
/v1/satellites	GET	JWT	List registered satellites and current state
/v1/satellites/{id}/telemetry	GET	JWT	Latest telemetry snapshot for satellite
/v1/commands	POST	JWT + MFA	Submit a telecommand for uplink
/v1/commands/{id}	GET	JWT	Get command status and ACK/NAK result
/v1/anomalies	GET	JWT	Active and historical anomaly alerts
/v1/orbital/{id}	GET	JWT	Current orbital state for satellite